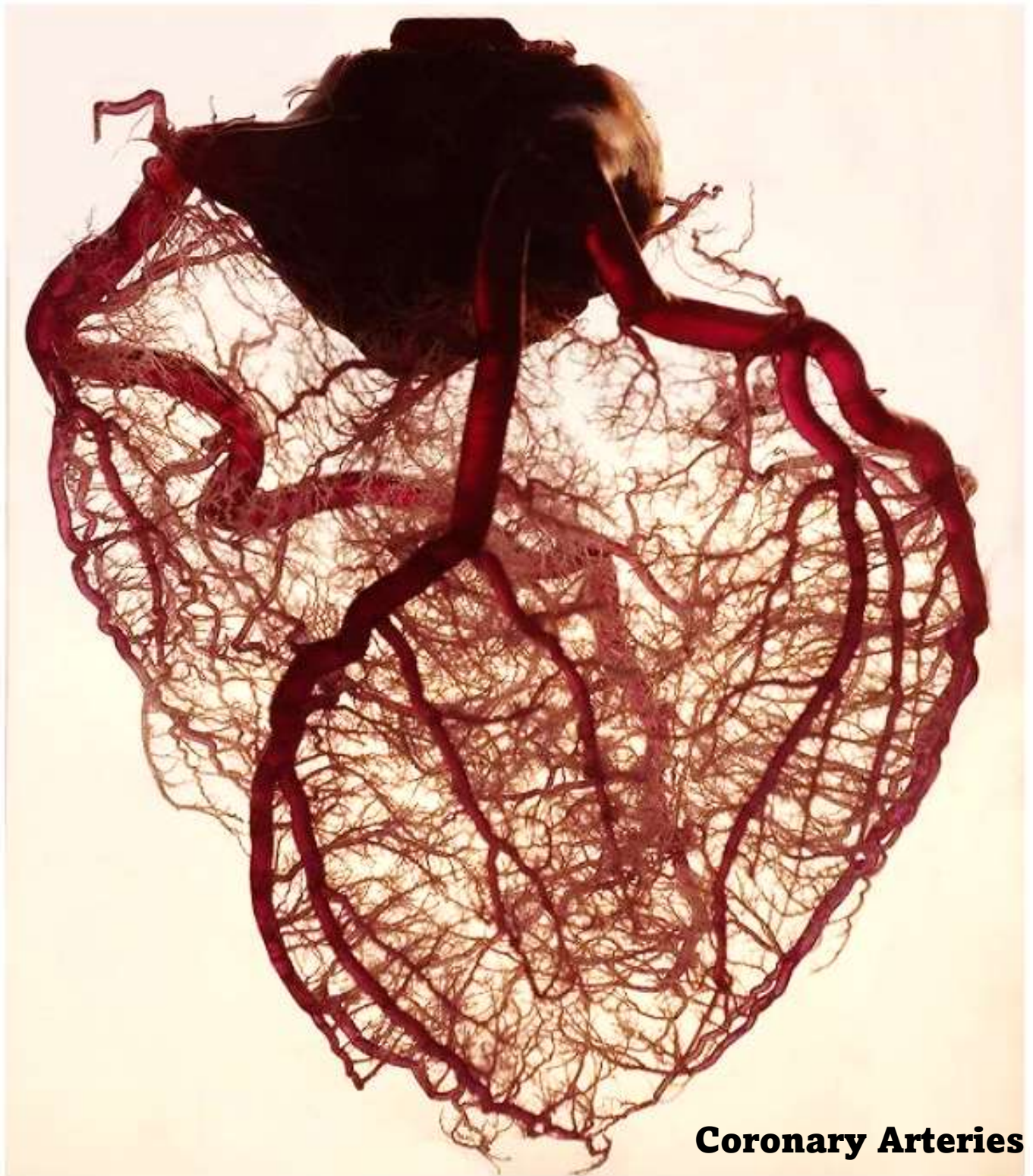
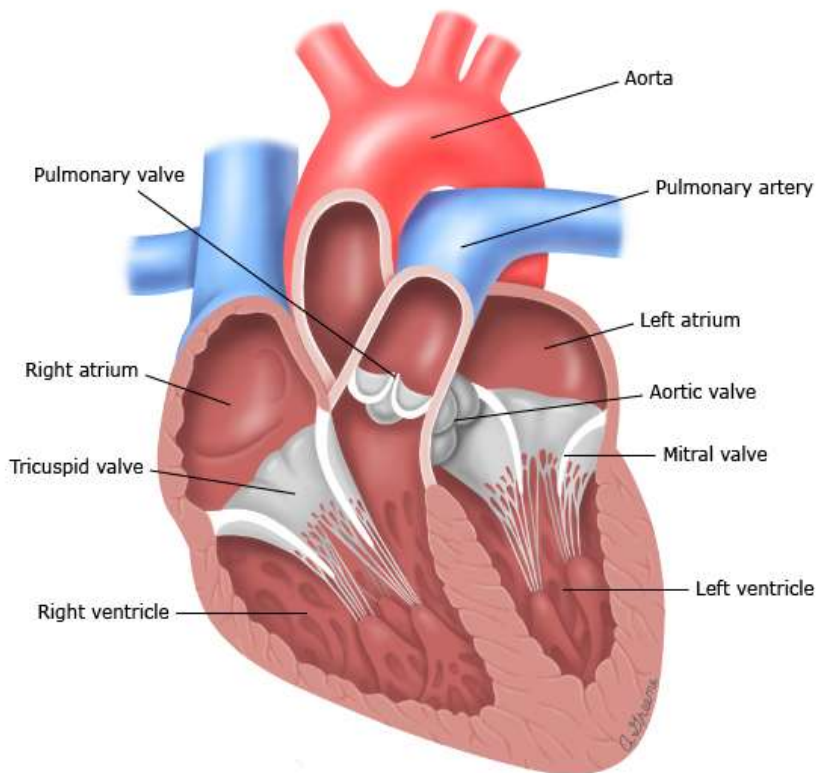


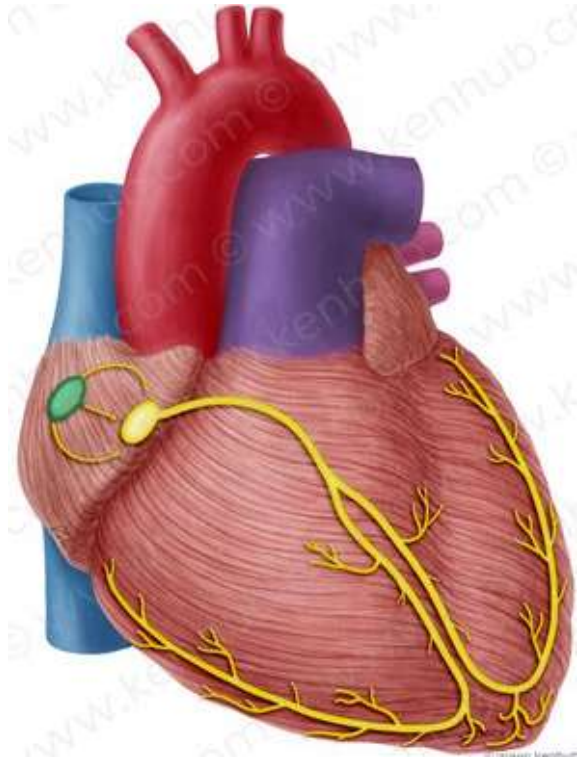
The Human Heart ❤️



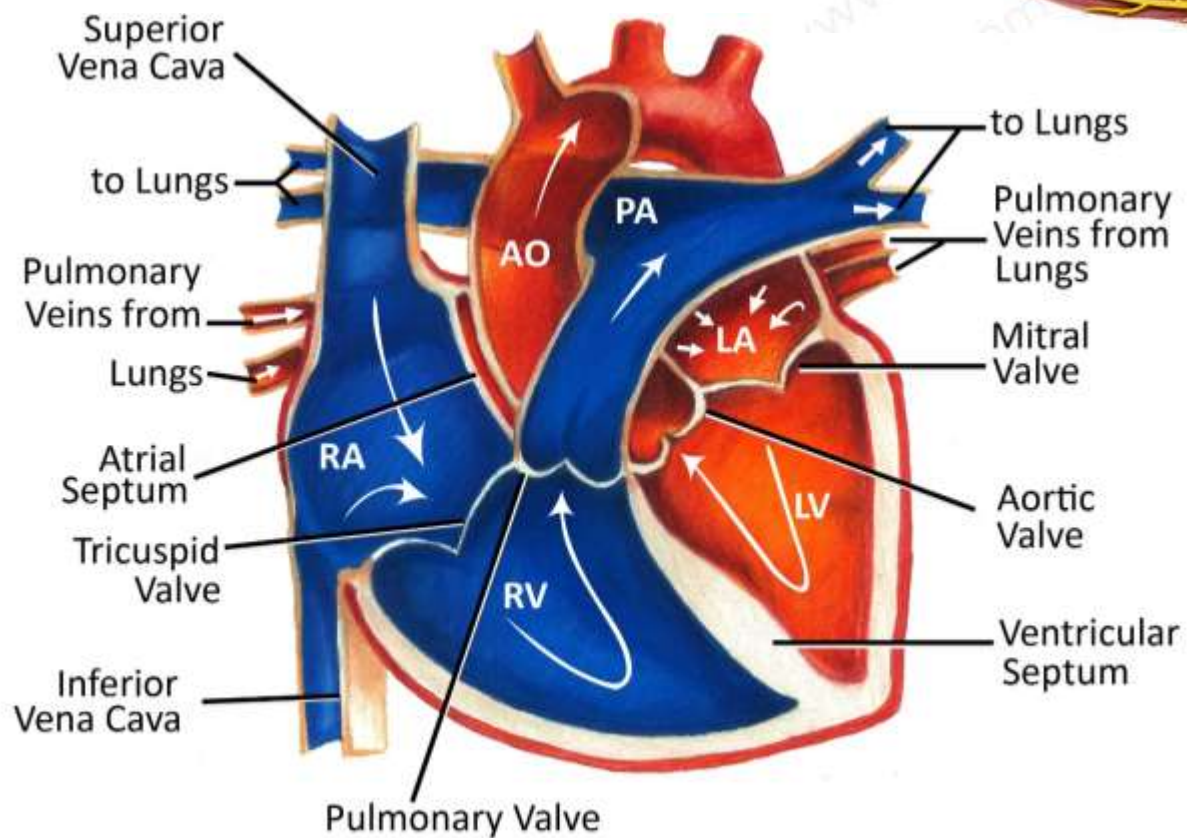
Coronary Arteries



Electricity of the heart

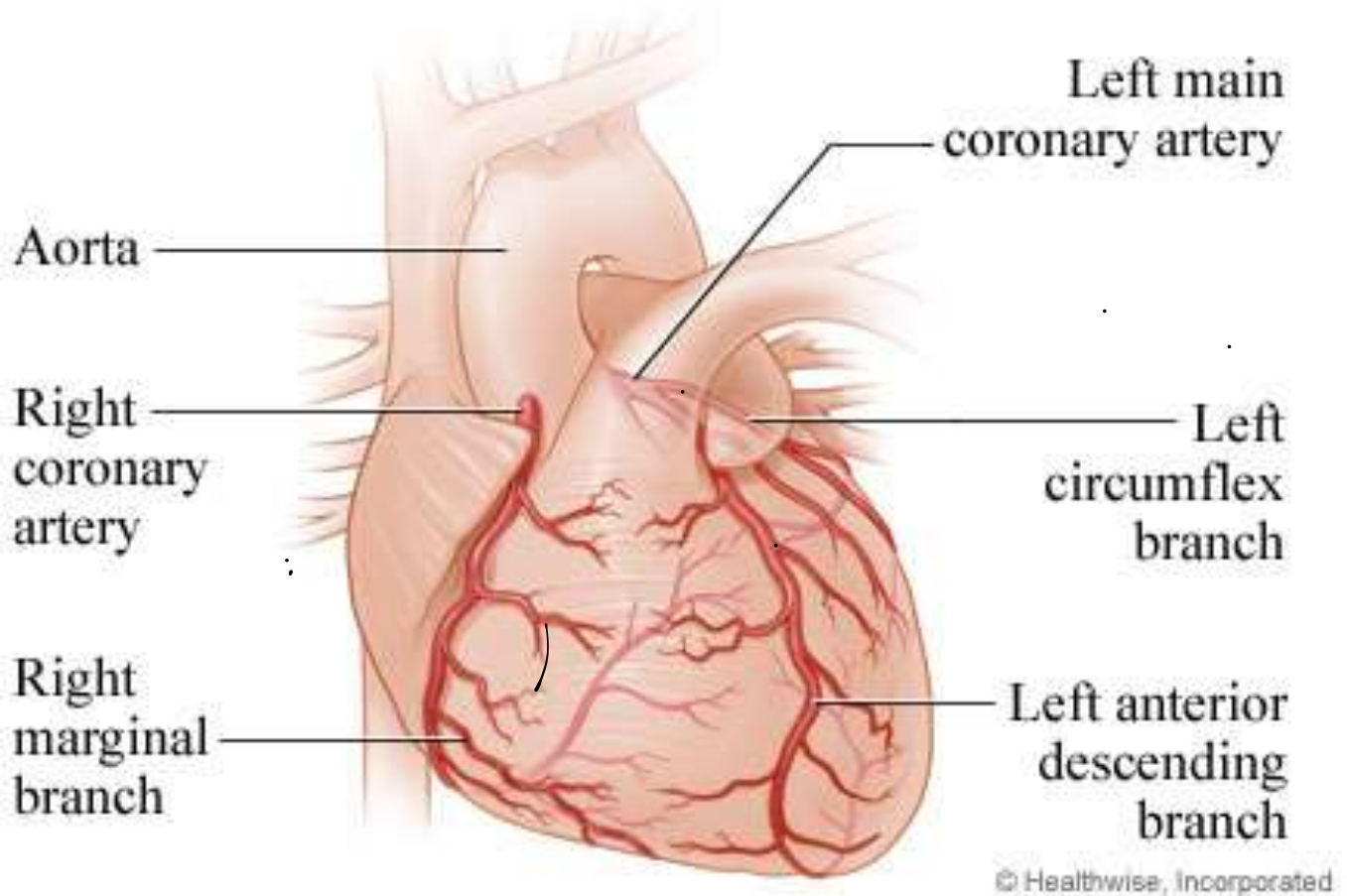


Blood flow THROUGH the heart



Coronary Artery Disease - CAD

**Blood flow TO the heart
via Coronary Arteries**



Coronary Artery Disease (CAD): pathological heart condition cause by plaque buildup (atherosclerosis) inside of the coronary arteries that decreases or restricts blood flow/oxygen delivery to the heart muscle.

RISK FACTORS FOR CAD

Non-modifiable (we cannot change these)	Modifiable (we can change these)
*Sex: Men > Women (risk ↑ after *menopause)	Hypertension
*Age (> 45)	Hyperlipidemia (high LDL, low HDL)
Family history/*genetics (especially parent with early onset CAD)	* Smoking
	Diabetes
Ethnicity (some groups have higher risks for example: South Asians & African Americans)	Physical inactivity/Obesity
	* Unhealthy diet (fats, salts, sugars)
	Stress & psychological factors
* = these factors contribute to increased cholesterol	

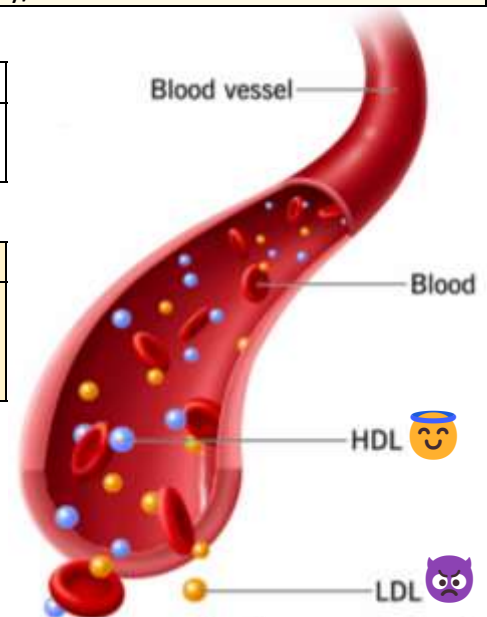
LIPID PROFILE (best obtained while fasting)

LDL Cholesterol "Low Density Lipoproteins" AKA: Low Quality Cholesterol	Less than 100mg/dL	Less than 70mg/dL for very high risk
LDL transports cholesterol and triglycerides into the cell and deposits on arterial vessel walls.		

HDL Cholesterol "High Density Lipoproteins" AKA: High Quality Cholesterol	Greater than 40mg/dL MALES	Greater than 50 mg/dL for FEMALES
HDL protects CVD by taking LDL away from the heart for the liver to process/remove from the body. HDL levels are increased through physical activity/exercise.		

Total Cholesterol	Less than 200mg/dL
Total Cholesterol clear predictor of coronary events. High levels ↑ risk of CVD regardless of age. Calculation = HDL + LDL + Triglycerides ÷ 5.	

Triglycerides	Less than 150mg/dL
Triglyceride levels increase after meals & are affected by stress. Diabetes, ETOH, and obesity cause elevation. Can lower HDL levels, increase plaque buildup, & very high levels cause pancreatitis.	

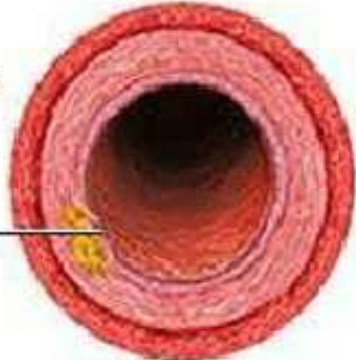


So what's happening exactly??

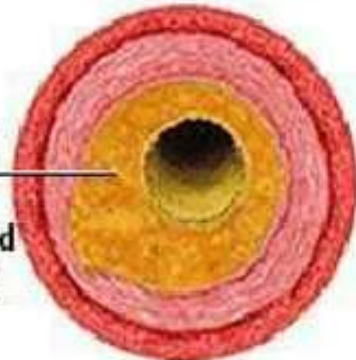
Normal
cut
section of
artery



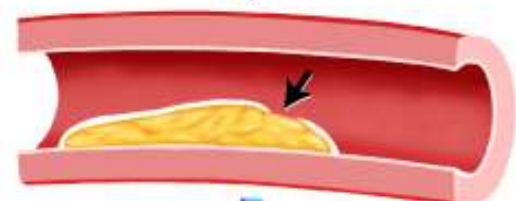
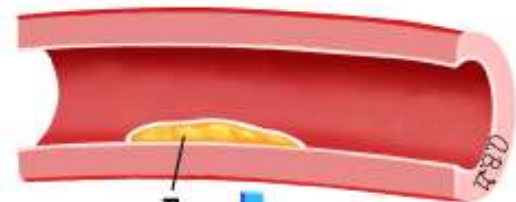
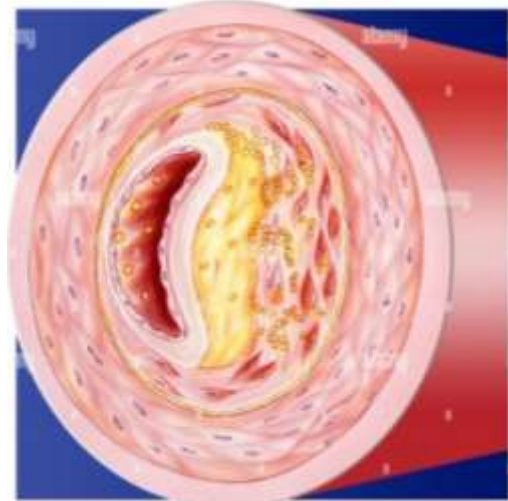
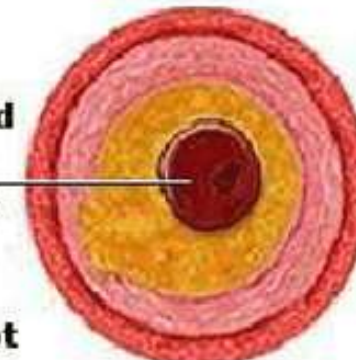
Tear in
artery
wall



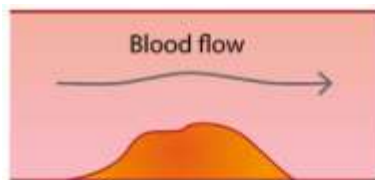
Fatty
material
is
deposited
in vessel
wall



Narrowed
artery
becomes
blocked
by a
blood clot



**VIEWS OF ARTERY:
Normal to Narrowed to
Completely Blocked**



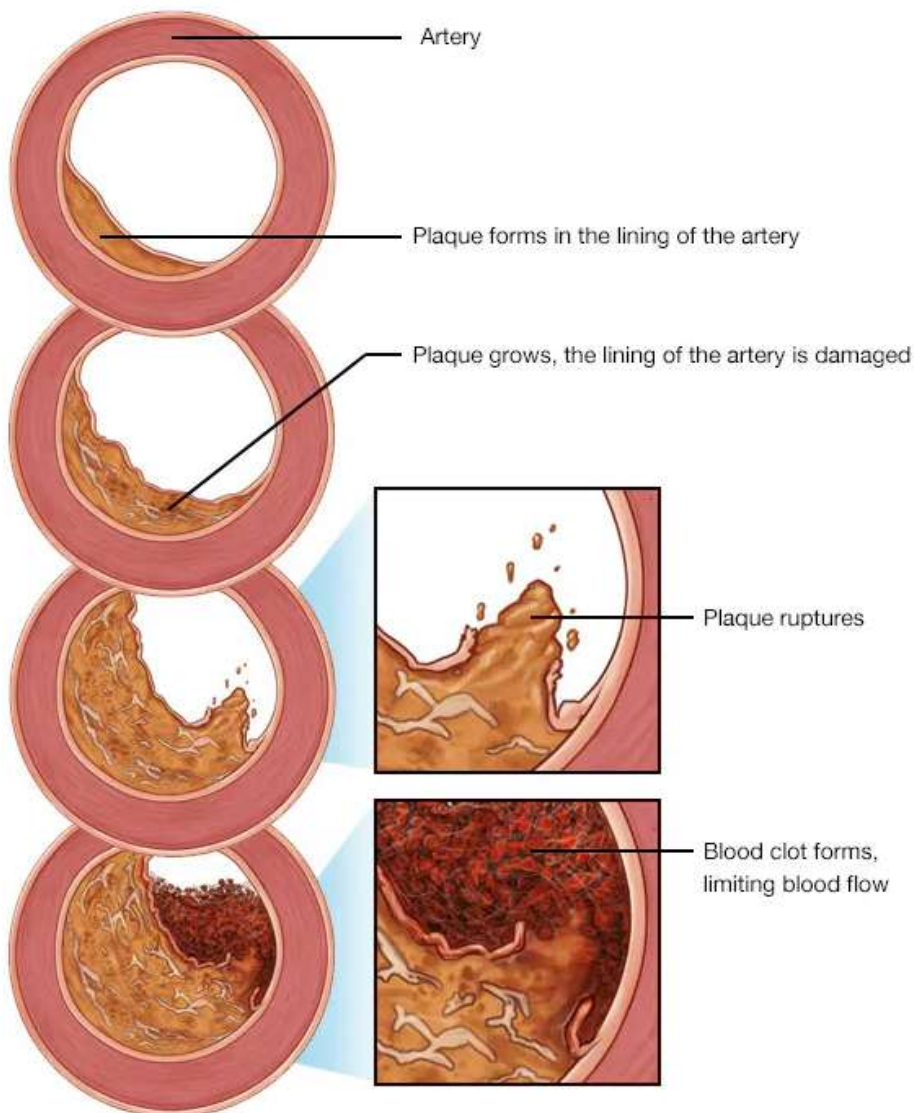
Coronary artery with atherosclerotic plaque. The core of the atherosclerotic plaque houses inflammatory cells and lipids which drive a chronic inflammation within the artery wall.



The atherosclerotic plaque is vulnerable (unstable). Rupture or erosion may damage its cap, which exposes highly thrombogenic materials. Such materials activate platelets and coagulation factors.



Activation of platelets and coagulation factors results in formation of a thrombus (atherothrombosis). Myocardium supplied by this artery becomes ischemic. Necrosis starts if flow is not restored within 30 minutes.



Art, the "meatball"

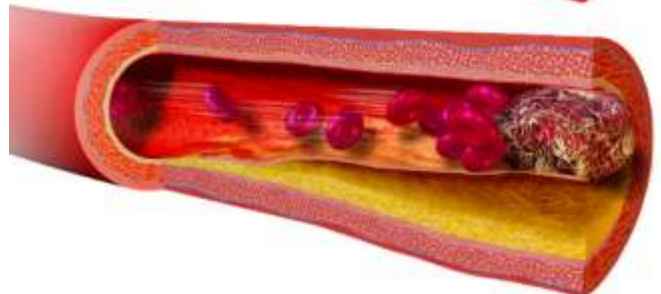
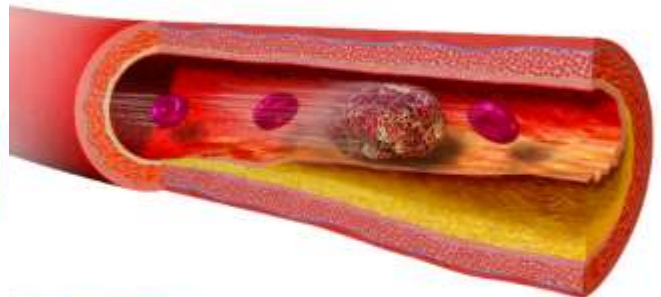
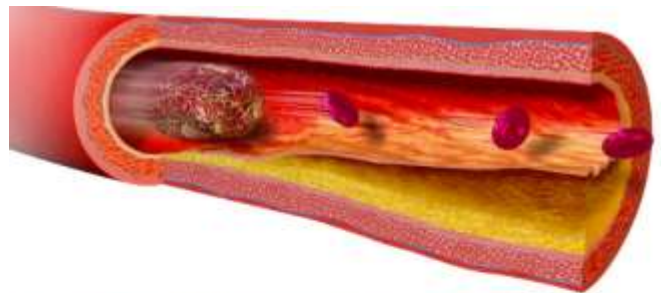
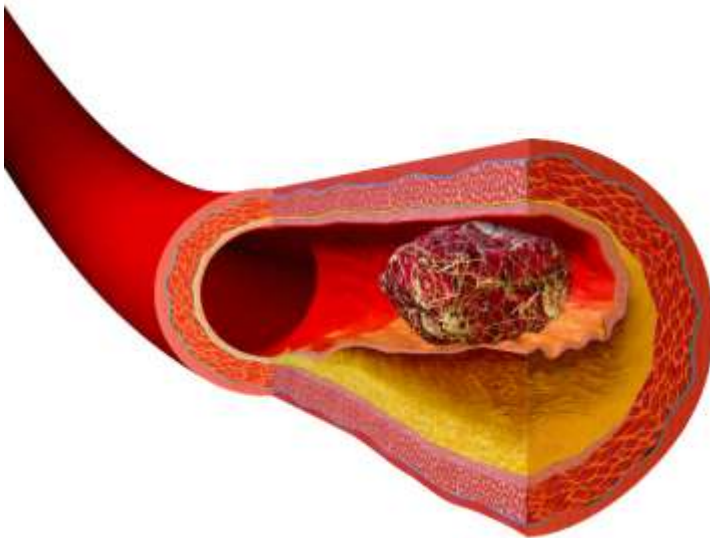
platelets & fibrin



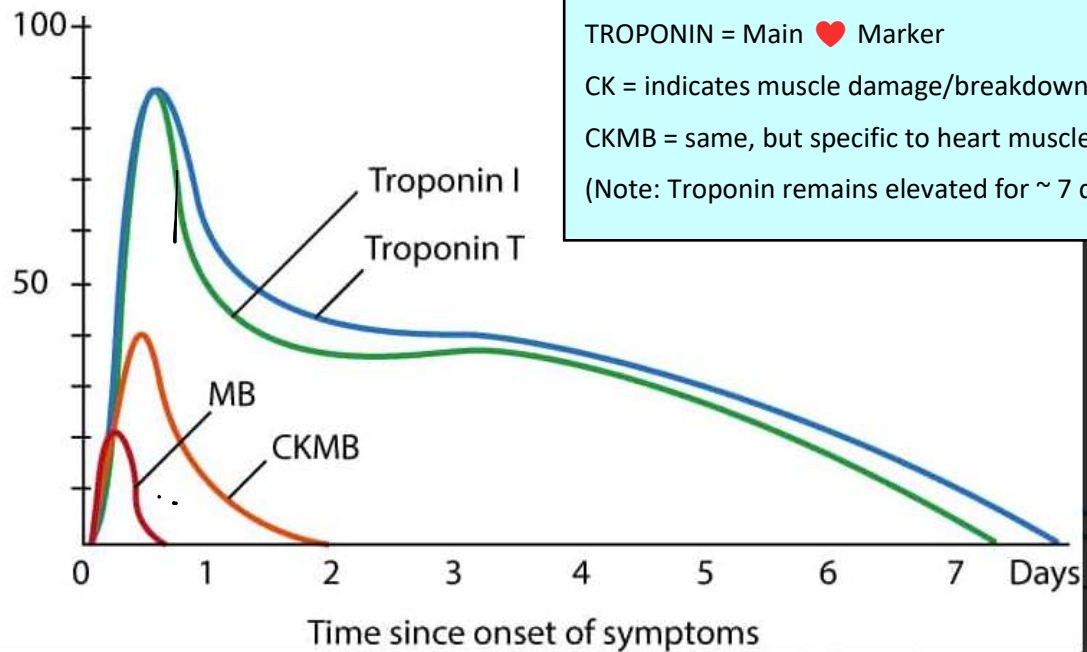
PATHOPHYSIOLOGY OF A BLOOD CLOT:

Intravascular Endothelial Injury > Activated Platelets gather > Clotting factors are released > Fibrin is created & comes in to hold everything together > More blood cells show up to "help" but keep collecting & a clot is formed & keeps growing larger.

Chrissy McCain, MSN, RN 04/2024



Blood levels of myocardial proteins
(times elevated over the reference limit)



Cardiac Markers/Enzymes

TROPONIN = Main ❤️ Marker

CK = indicates muscle damage/breakdown

CKMB = same, but specific to heart muscle

(Note: Troponin remains elevated for ~ 7 days)

ANGINA

Chest Pain, Pressure, or Discomfort; Symptom of lack of oxygen to heart muscle. Can be midsternal, squeezing, burning, tightness, can radiate to arms, neck, jaw, back, or mimic indigestion pains (especially in women).

Stable: *Provoked by activity & Able to be relieved with rest or medication*

Unstable: *Happens even at rest, UNable to be relieved with rest or meds, increasing in severity*

Note: you can have Angina without an MI. Many pt's with chronic stable angina take Nitro at home or long-acting nitrates daily.

**Assessing for Acute Coronary Syndrome or Acute Myocardial Infarction**

Be alert for the following signs and symptoms:

Cardiovascular

- Chest pain or discomfort not relieved by rest or nitroglycerin; palpitations. Heart sounds may include S₃, S₄, and new onset of a murmur.
- Increased jugular venous distention may be seen if the myocardial infarction (MI) has caused heart failure.
- Blood pressure may be elevated because of sympathetic stimulation or decreased because of decreased contractility, impending cardiogenic shock, or medications.
- Irregular pulse may indicate atrial fibrillation.
- In addition to ST-segment and T-wave changes, the electrocardiogram may show tachycardia, bradycardia, or other arrhythmias.

Respiratory

Shortness of breath, dyspnea, tachypnea, and crackles if MI has caused pulmonary congestion. Pulmonary edema may be present.

Gastrointestinal

Nausea, indigestion, and vomiting.

Genitourinary

Decreased urinary output may indicate cardiogenic shock.

Skin

Cool, clammy, diaphoretic, and pale appearance due to sympathetic stimulation may indicate cardiogenic shock.

Neurologic

Anxiety, restlessness, and lightheadedness may indicate increased sympathetic stimulation or a decrease in contractility and cerebral oxygenation. The same symptoms may also herald cardiogenic shock.

Psychological

Fear with feeling of impending doom, or denial that anything is wrong.

Patients experiencing MI will present in many ways with different types of chest pain; vitals/labs/EKG will vary among patient having the same type of heart attack.

CENTRAL ILLUSTRATION: Patient Assessment With Suspected ACS					
	LOW ————— Likelihood of myocardial infarction (MI) ————— HIGH				
I. Clinical setting Symptoms and vital signs					 CPR/shock
II. Electro-cardiogram (ECG)	 Normal ECG	 ST depression (mild)	 ST depression	 ST elevation	
III. Troponin level at 0h		—	—/+	+	++ +++
IV. Troponin change (within 1, 2 or 3h)		—	—/+	+	++ If any of the above, consider direct rule-in
Triage decision	Rule-out MI		Observe		Rule-in MI
Differential diagnosis	Noncardiac		Unstable angina	Other cardiac	NSTEMI STEMI

We must consider all these assessment and diagnostic findings together in determining if a patient is having an MI.

Acute Management of Suspected MI:

- Assessment
- EKG & Evaluate
- Lab work
- IV Heparin
- Cath Lab
- **MONA – ONAM – MOAN – ANOM – MAON – NMAO – OMAN – ONMA**

Order DOES NOT MATTER!

Aspirin – Nitro – Oxygen – Morphine

**MONA &
expect an IV
Heparin drip**

Blood Thinners AKA Anticoagulants / Anticoags / ACs

ANTICOAGULANTS

Brand Name	Chemical Name	Mechanism of Action	Indication(s)
Coumadin	Warfarin	Works on clotting cascade in different areas	Prevention of “peripheral clots” and clots inside of the heart “with Afib” and mechanical valve replacements.
Heparin	Heparin		
Eliquis	Apixaban		
Pradaxa	Dabigatran		
Xarelto	Rivaroxaban		

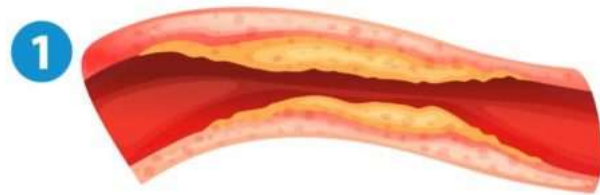
ANTI-PLATELET AGGREGATES

Brand Name	Chemical Name	Mechanism of Action	Indication(s)
Brilinta	Ticagrelor	Inhibits platelet aggregation for the lifespan of the platelet to prevent clots.	Prevention of arterial clots in MI, established PAD, sometimes stroke (higher bleeding risk pts).
Effient	Prasugrel		
Plavix	Clopidogrel		
Pletal	Cilostazol		

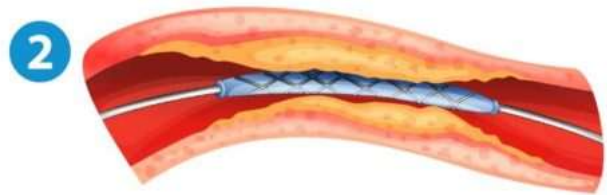
Know the difference between Antiplatelet Aggregates & Anticoagulants and when/why they're indicated, monitoring, pt. education.

Procedure / Surgical Interventions

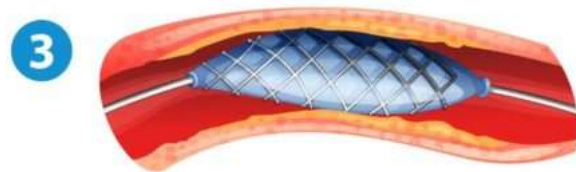
Stent with Balloon Angioplasty



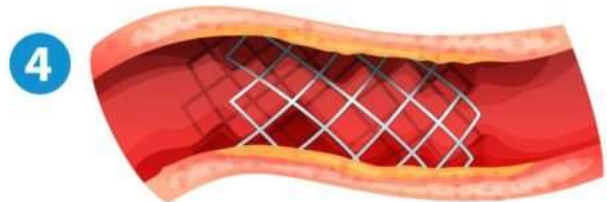
1 Build up of cholesterol partially blocking blood flow through the artery.



2 Stent with balloon inserted into partially blocked artery.

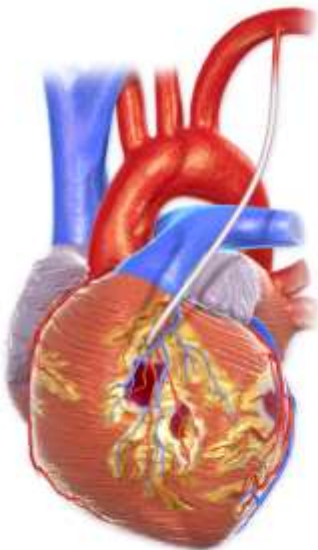


3 Balloon inflated to expand stent.

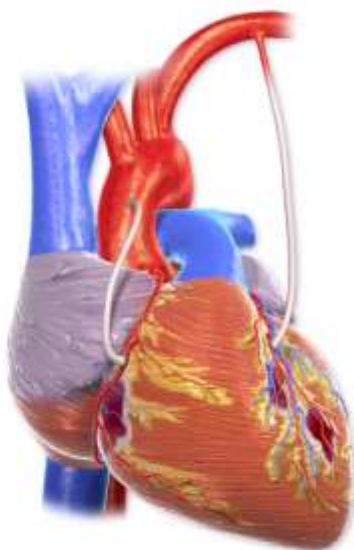


4 Balloon removed from expanded stent.

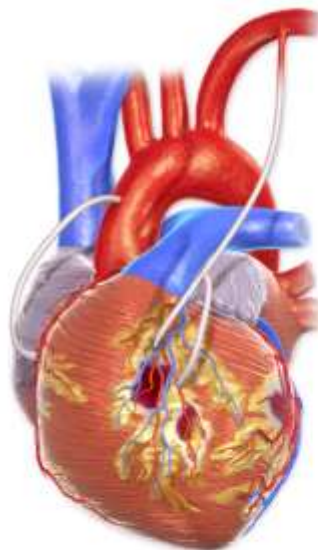
Single



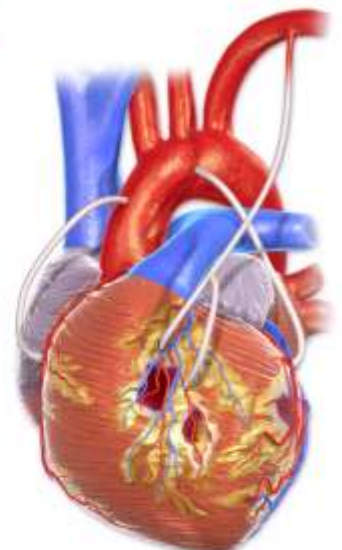
Double



Triple



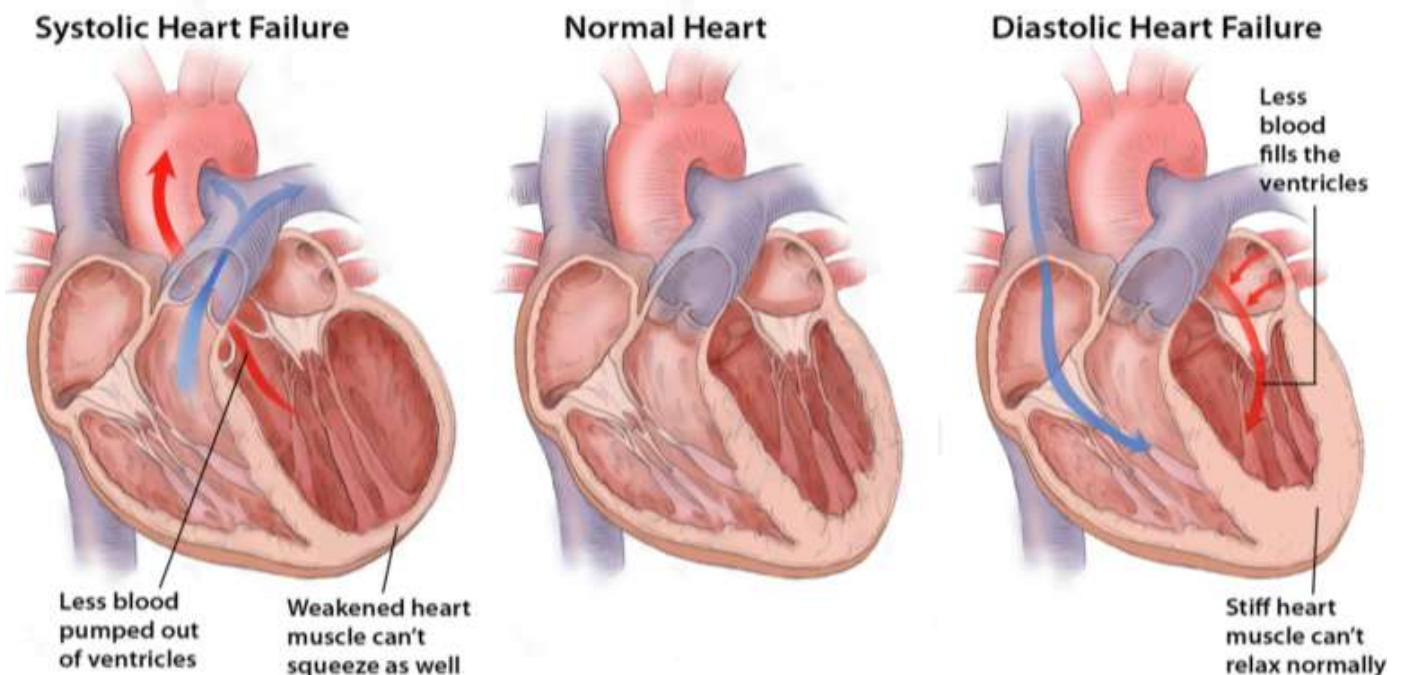
Quadruple



Coronary Artery Bypass Graft (CABG)

Heart Failure

"Heart failure (HF) is a clinical syndrome resulting from structural or functional cardiac disorders so that the heart is unable to pump enough blood to meet the body's metabolic demands or needs (American Heart Association [AHA], 2019a). The term heart failure indicates myocardial disease in which impaired contraction of the heart (systolic dysfunction) or filling of the heart (diastolic dysfunction) may cause pulmonary or systemic congestion."



- **SYSTOLIC**

- Weakened heart muscle unable to efficiently pump blood to the rest of the body. Fluid backs up to the lungs then the right side of the heart.

- **DIASTOLIC**

- Impaired filling of left ventricle due to stiffened or hypertrophy in myocardium

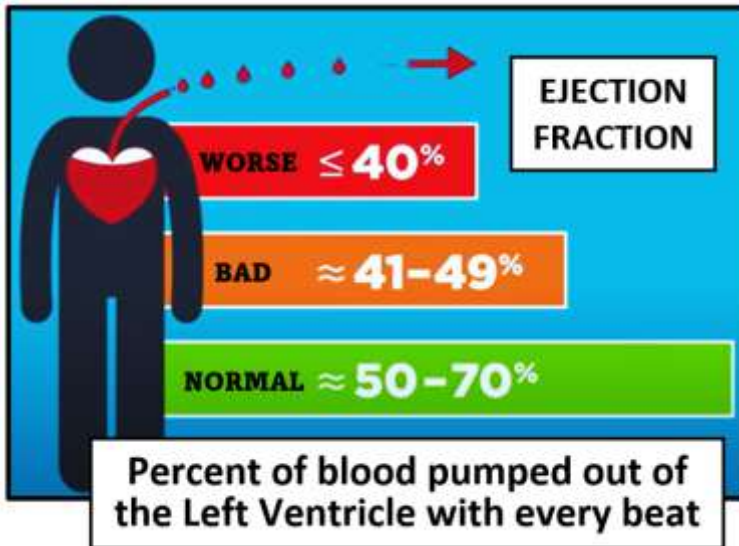
- **LEFT-SIDED**

- Effects left side of heart. 😊 Fluid backs up to lungs

- **RIGHT-SIDED**

- Effects right side of heart. 😊 Fluid backs up to the systemic circulation

ECHOCARDIOGRAM: Ultrasound test to look at the muscle contraction & ability of the heart including measuring the Ejection Fraction. It also looks at the components of the heart like valves and can identify structural deformities.



SYSTOLIC vs. DIASTOLIC HEART FAILURE

SYSTOLIC:

"HFrEF" (sounds like "heff reff")
Heart Failure with Reduced Ejection Fraction
PUMPING ISSUE that Reduces or Lowers the EF.

DIASTOLIC:

"HFpEF" (sounds like "heff peff")
Heart Failure with Preserved Ejection Fraction
FILLING ISSUE that doesn't affect or preserves the EF.

LEFT SIDE vs RIGHT SIDE
— is not —
the same as SYSTOLIC vs DIASTOLIC



What you need to know:

SYSTOLIC vs DIASTOLIC = How is the Left Ventricle affected???

Systolic = weakened muscle, impaired LV CONTRACTION

Diastolic = thick, stiffened muscle, impaired FILLING

LEFT vs RIGHT = Which side of the heart is affected???

Left-sided = Left side so it backs up to the lungs (to the right)

Right-sided = Right side so it backs up to venous circulation

RIGHT SIDED ♥ FAILURE

(Cor Pulmonale)

- Fatigue
- ↑ Peripheral Venous Pressure
- Ascites
- Enlarged Liver & Spleen
- May be secondary to chronic pulmonary problems
- Distended Jugular Veins
- Anorexia & Complaints of GI Distress
- Swelling in Hands & Fingers
- Dependent Edema



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LEFT SIDED ♥ FAILURE

- Paroxysmal Nocturnal Dyspnea
- Elevated Pulmonary Capillary Wedge Pressure
- Pulmonary Congestion
 - Cough
 - Crackles
 - Wheezes
 - Blood-Tinged Sputum
 - Tachypnea
- Restlessness
- Confusion
- Orthopnea
- Tachycardia
- Exertional Dyspnea
- Fatigue
- Cyanosis



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Left-Sided vs. Right-Sided Heart Failure

Right-Sided

Heart loses some of its ability to move oxygen-depleted blood to the lungs to pick up new oxygen

Most often caused by left-sided heart failure

Can also occur even when the left side of the heart is apparently normal



Left-Sided

Most common type of heart failure

Heart loses some of its ability to pump blood out to your body after it's been re-oxygenated

Usually caused by coronary artery disease



Congestive Heart Failure



Triggers or Precipitating Factors

- Infection or Illness
 - Influenza, respiratory, etc.
- Infective Endocarditis
 - Infection of heart
- Mitral Valve Regurgitation
- Myocardial Ischemia
 - Decreased blood flow to heart
- Myocardial Infarction
 - Heart attack

- Cardiac Arrhythmias
 - Abnormal heart rate/rhythm
- Pulmonary Embolism (PE)
 - Blood clot in lung artery
- Anemia
- Thyrotoxicosis
- Pregnancy
- Medications/Non-compliance
- Alcohol Consumption



Congestive Heart Failure



Treatment

General Measures

- Treat underlying cause or precipitating factors
- Diet modifications
 - Sodium restriction
 - Possible fluid restriction
- Adequate rest
- Mild exercise
- Alcohol/Smoking cessation
- Adequate oxygen
- Health maintenance
 - Vaccines, wellness checks

Medications

- Diuretics
- ACE Inhibitors
- Angiotensin Receptor Blockers (ARBs)
- Beta Blockers
- Cardiac Glycosides

Surgical

- Revascularization
 - Angioplasty, CABG
- Valve Repair/Replacement

Device Therapy

- Cardiac Resynchronization (CRT)
 - Biventricular pacing
- Implantable Cardioverter Defibrillator (ICD)
 - To reduce incidence of cardiac death seen
- Ventricular Assisted Devices (VAD)
 - Often used as a bridge while waiting for definitive therapy



Congestive Heart Failure



Treatment - Medications

Class	Action	Medications	Oral Dose	Side Effects
Angiotensin Converting Enzyme Inhibitor (ACEI)	Inhibit conversion of Angiotensin I to Angiotensin II	Captopril Lisinopril Enalapril Ramipril Perindopril	6.25-25 mg TID 2.5-10 mg OD 2.5-10 mg BID 1.25-10 mg OD 4-8 mg OD	Hypotension, cough, aggravation of renal failure, hyperkalemia, angioedema
Angiotensin Receptor Blocker (ARB)	Block action of Angiotensin II on Angiotensin II Receptors	Losartan Valsartan Candesartan	12.5-25 mg BID 40-80 mg BID 4-32 mg OD	Hypotension, angioedema, aggravation of renal failure, hyperkalemia
Beta Blockers (BB)	Block Alpha, Beta-1 & Beta-2 Receptors	Carvedilol Bisoprolol Metoprolol (SR)	3.125-25 mg BID 1.25-25 mg OD 12.5 – 200 mg OD	Bradycardia, heart blocks, fatigue, aggravation of asthma, low BP
Digoxin	Positive inotropic effect. Inhibition of AV node by increased vagal tone	Digoxin	0.125-0.25 mg OD	Cardiac arrhythmias, visual disturbances. GI disturbances

★**DIGOXIN**★ Digoxin increases the force of myocardial contraction and slows conduction through the AV node. It improves contractility, increasing Left ventricular output. Although the use of digoxin does not result in decreased mortality rates among patients with HFrEF, it can be effective in decreasing the symptoms of HF and may help prevent hospitalization.

Digoxin Toxicity aka Dig & Digitalis toxicity – S&S: weakness, tiredness, anorexia, nausea, visual disturbances, confusion, and bradycardia or other arrhythmias especially in setting of HYPOmagnesemia. Monitor K⁺ because the effect of digoxin is enhanced in the presence of HYPOkalemia and digoxin toxicity may occur. A serum digoxin level is obtained if the patient's renal function changes or there are symptoms of toxicity. Also contributes to HYPERcalcemia especially when taken with ACE-Inhibitors, ARBS, or K-sparing diuretics (spironolactone)

- MedSurg Book, HF Chapter

We check daily weights on patients because it is the best indicator of fluid status. We measure strict intake/output. If unable to measure urine output, weigh briefs or chucks pads. We also manage fluid restrictions closely in acute care.

DC TEACHING: Patients should follow fluid restriction as ordered, weigh themselves daily, check lower extremities for edema, and monitor for S&S of decompensation.

When to call the provider:

Weight gain of 1-3lbs in a day or 3-5 lbs in 3 days
New/worsening lower extremity edema
New/worsening SOB

Peripheral VASCULAR disease

Arteries	Veins	
PAD Peripheral Artery Disease	CVI Chronic Venous Insufficiency	PVD Peripheral Venous Disease

Peripheral VASCULAR Disease

PERIPHERAL ARTERY DISEASE

ARTERIAL NARROWING OR OCCLUSION (ARTERIOSCLEROSIS) WHICH CAUSES DECREASED O₂ AND NUTRIENTS TO THE LOWER EXTREMITIES.

PERIPHERAL ARTERY DISEASE

1. PAIN (SHARP)
2. ABSENT PULSE
3. SKIN:
 - A. COOL TO TOUCH
 - B. PALE SKIN
 - C. ABSENT HAIR • SHINY SKIN
 - D. THIN, DRY • SCALY SKIN
 - E. NO EDEMA
4. LESIONS:
 - A. RED SORES ON THE TOES/FEET
 - B. PUNCHED OUT APPEARANCE
5. GANGRENE (DEATH OF TISSUES)



CHRONIC VENOUS INSUFFICIENCY

POOLING OF BLOOD IN THE EXTREMITIES DUE TO THE INABILITY TO BRING BLOOD BACK TO THE HEART (VASCULAR INSUFFICIENCY)

CHRONIC VENOUS INSUFFICIENCY

1. PAIN (ACHY • DULL)
2. PRESENCE OF A STRONG PULSE
3. SKIN:
 - A. PRESENCE OF EDEMA
 - B. WARM LEGS
 - C. YELLOW/BROWN ANKLES
4. LESIONS:
 - A. IRREGULAR SHAPED SORES
5. NO PRESENCE OF GANGRENE



PERIPHERAL ARTERY DISEASE

1. HANG (DANGLE) THE PATIENT'S LEGS



CHRONIC VENOUS INSUFFICIENCY

1. ELEVATE THE PATIENT'S LEGS



PERIPHERAL ARTERY DISEASE

- Assess extremities thoroughly
- Provide skin care/avoid soaking
- Avoid constriction: Don't cross legs or wear compression stockings
- Manage pain/administer meds

• Patient Education:

- Teach foot care/wear warm socks
- Inform client of disease & s/s of ↓ in circulation
- Teach about optimal diet/exercise
- Teach client to relieve extremity pain by placing limbs in a dependent position. Don't elevate above heart

Nursing Implications/
Patient Education

Collaborative Care

• Risk modification:

- Smoking/Tobacco cessation
- Management of DM/HTN
- Control lipid levels/↓ weight
- Exercise/Nutritional Therapy
- Inspect feet regularly

• Drug Therapy:

- Antiplatelets
- Statins
- ACE Inhibitors

• Procedures/Surgery

- Bypass graft
- PCI/stents
- Atherectomy

* Peripheral arteries become narrowed and thickened

- Leading cause: atherosclerosis

Pathophysiology

Risk Factors

- Tobacco use
- Diabetes
- Hypertension
- Hyperlipidemia
- Obesity
- Family history

Peripheral Artery Disease (PAD)

Assessment/Clinical Manifestations

Lab/Diagnostics

- Duplex doppler imaging/doppler pulse readings
- Segmental BPs
- Ankle-brachial index
- Lipid panel/CRP
- Blood glucose levels (DM)

Complications

- May progress to rest pain
- Ulceration
- Gangrene/infections
- Amputations

• Intermittent claudication:

- Ischemic muscle pain usually caused by exercise.

• Paresthesia: numbness/tingling

• Physical appearance:

- Pulses ↓ or absent
- Prolonged cap refill
- Cool extremities
- Skin: shiny, thin
- Hair loss
- Thickening of toenails
- Elevation pallor/dependent rubor
- Ulcers: round, smooth black eschar/pale pink granulation

Ankle-Brachial Index "ABI":

Quick check for PAD (Peripheral Artery Disease)

HOW TO MEASURE:

Divide the Ankle Systolic Pressure (SBP)
by the
Brachial Systolic Pressure (SBP)

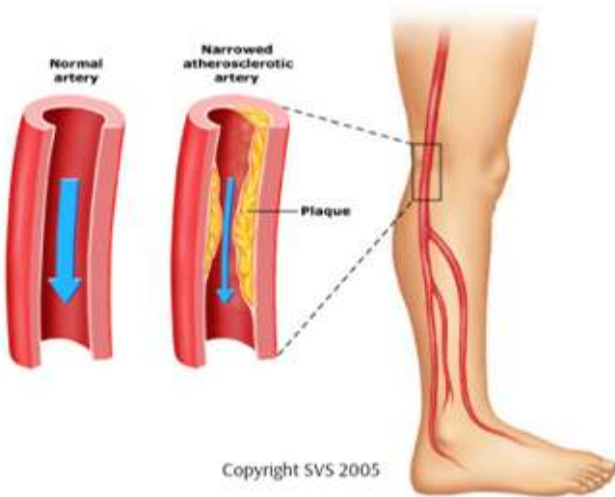
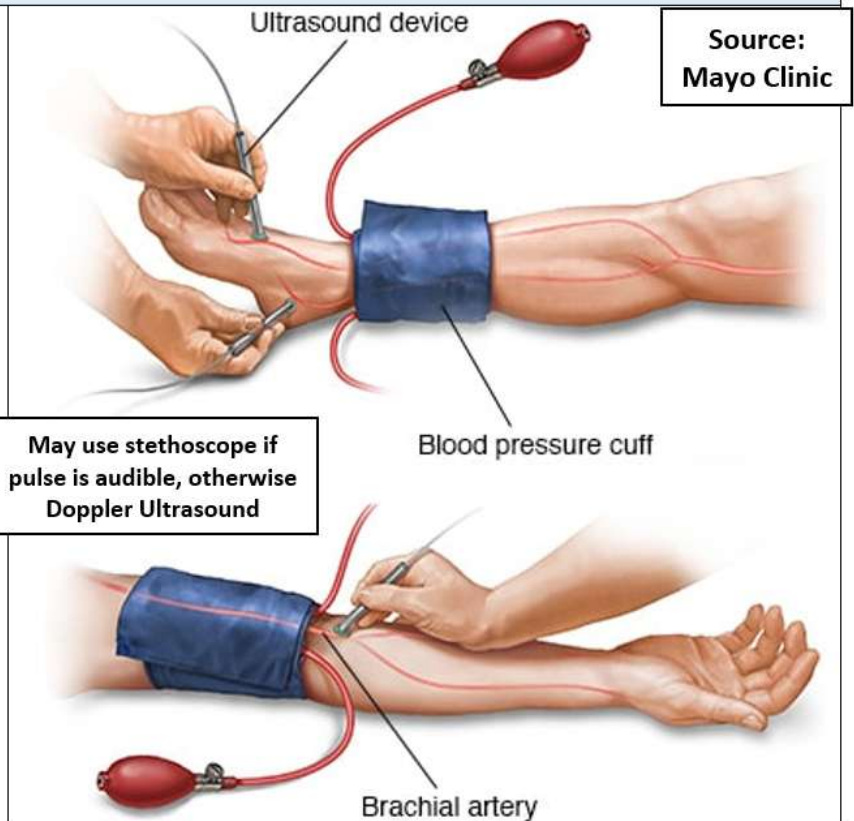
AKA:

$$\text{Ankle SBP} / \text{Arm SBP} = \text{ABI}$$

- 1.0 to 1.4 = No Blockage or Narrowing identified or suspected
- 0.90 to 0.99 = Indicated borderline PAD

Less than 0.90 = Indicates a Dx of PAD & more testing is needed/likely to confirm/evaluate further.

Reference: ABI Med Surg Book - Ch 26, pg. 817/823



Intermittent Claudication

Claudication, also referred to as intermittent claudication, is pain caused by a reduced blood flow to a limb generally in the legs. Claudication is a symptom of a disease rather than a disease itself. Most often claudication is a symptom of peripheral artery disease or PAD.

Symptom of Claudication:

- Pain in calves, thighs, feet, hips or buttocks
- Pain in calves, legs, etc... when at rest
- Discolored skin, bluish or cold to the touch
- Ulcerations or sores on feet, lower legs, toes or fingers
- Weakness, Aching or burning sensation

Causes of Claudication:

Claudication is generally a symptom of peripheral artery disease as a result of atherosclerosis or a stiffening & hardening of the arteries due to a build up of plaque. This build up of plaque narrows the artery making it difficult for oxygen rich blood to flow to the outer limbs. The lack of oxygen rich blood flowing to the muscles, because of plaque, causes this claudication pain in the limb.

Diagnosis & Treatment:

Claudication can be diagnosed by a Vascular Surgeon through testing such as an ABI.

Treatment options include medication management, Angioplasty, or replacing the damaged vessel with a healthy vessel from another part of your body.

CHRONIC VENOUS INSUFFICIENCY

- Assess wounds/ulcers for signs of infection
- Daily moisturizers for eczema
- Teach client how to correctly apply compression stockings
- Teach proper skin care
- Teach client to avoid staying in one position for long periods; leg exercises
- Teach client to elevate their legs (above heart)

Nursing
Implications/
Patient Ed.

- * Usually caused by varicose veins or complication of DVT
- * Fluid/RBCs leak into tissues = Brown discoloration

pathophysiology

Risk Factors

- History of DVTs
- Age
- Obesity
- pregnancy
- sitting/standing for long periods

Assessment/Clinical Manifestations

Chronic Venous
Insufficiency

Collaborative
Care

Labs/Diagnostics

Complications

Assess skin/circulation:

- Lower leg skin: leathery
- Brownish appearance
- Edema present
- Eczema/itching
- painful venous ulcers
- Adequate cap refill
- skin warm
- Pulses present; may be difficult to palpate due to edema
- ulcers: irregular shape, yellow slough or ruddy granulation
- dull ache pain in leg

- Duplex ultrasound with Doppler
- CT venogram/venogram

- Venous ulcers
- osteomyelitis
- malignant changes
- Amputations (rare)

3E's

- Elevation
- Exercise
- Elastic stockings

Moist dressing

Nutritional Therapy:

- High in protein
- vitamins/supplements
- Tight glucose control for DM patients

Antibiotics for wound infection

- Systemic antibiotics
- should be guided by wound culture results
- wound debridement and excision

